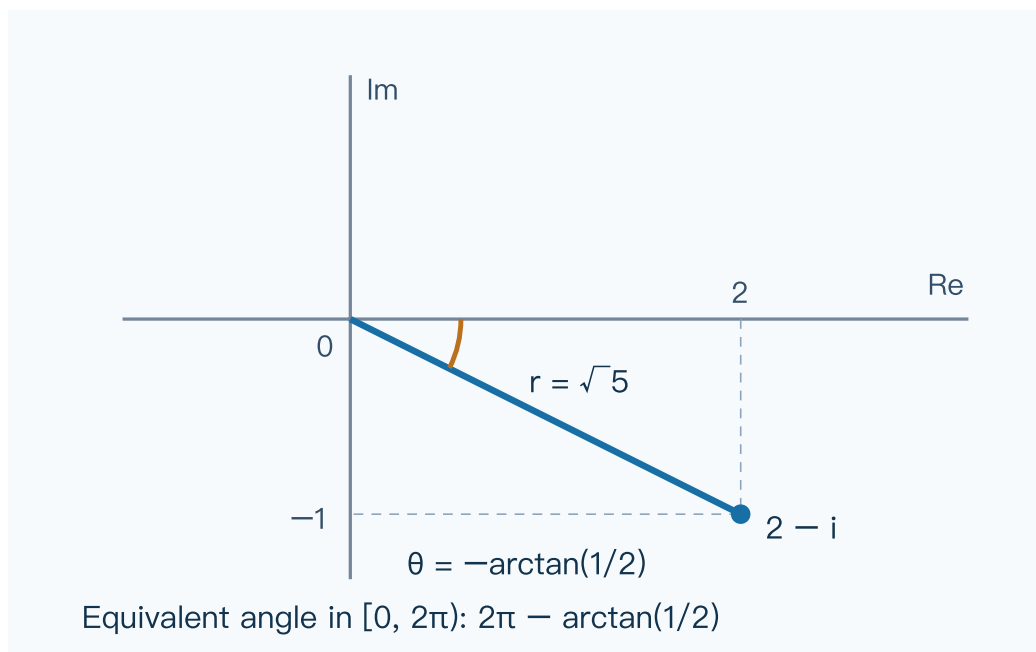


A1 · Warm-up: representing $2 - i$ · 热身：同一复数的不同表示

Linear Algebra · Week 1 · 25 September 2026

TEACHER ONLY — contains solutions | 先让学生尝试，再按顺序画图与补答案

How many different ways can we represent the complex number $2 - i$?



$$2 - i \leftrightarrow (2, -1), \quad r = \sqrt{5}, \quad \theta = -\arctan(1/2)$$

Board sequence · 板书顺序

1. Draw horizontal Re and vertical Im axes. Mark 2 on Re and -1 on Im; plot their intersection.
先画 Re 横轴、Im 纵轴；标出 2 和 -1，再画交点。
2. Join the origin to the point. Add dashed projections and label the length $\sqrt{5}$ only after students calculate it.
连原点到该点；画虚线投影。学生算完后才标长度 $\sqrt{5}$ 。
3. Mark the short clockwise angle $-\arctan(1/2)$. If an angle in $[0, 2\pi)$ is required, write $2\pi - \arctan(1/2)$.
先标顺时针负角；若要求 $[0, 2\pi)$ ，再写等价正角。不要一开始抄满答案。

Ask before explaining: Why is a positive acute angle wrong for this point?

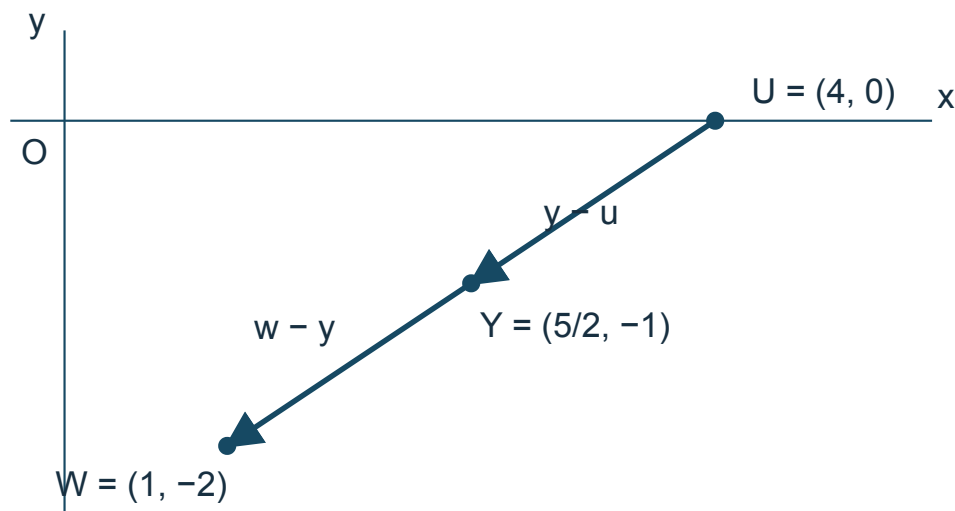
正锐角给正虚部，但原数虚部为负；点必须在第四象限。

A2 · Vector calculations and a midpoint · 向量运算与中点

Linear Algebra · Week 1 · 25 September 2026

TEACHER ONLY — contains solutions | 先让学生尝试，再按顺序画图与补答案

Let $\mathbf{u} = \begin{pmatrix} 4 \\ 0 \end{pmatrix}$, $\mathbf{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, $\mathbf{w} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$. Find $\mathbf{x}, \mathbf{y} \in \mathbb{R}^2$ where $\mathbf{x} = 3\mathbf{u} - 2(\mathbf{v} + \mathbf{w})$ and $\mathbf{w} - \mathbf{y} = \mathbf{y} - \mathbf{u}$.



$U \rightarrow Y$ and $Y \rightarrow W$: equal displacement, same direction.

$$\mathbf{x} = (8, 0), \quad 2\mathbf{y} = \mathbf{u} + \mathbf{w}, \quad \mathbf{y} = (5/2, -1)$$

Board sequence · 板书顺序

1. For the first calculation, write $\mathbf{v} + \mathbf{w} = (2, 0)$, then $\mathbf{x} = (12, 0) - (4, 0) = (8, 0)$. Keep this separate from the midpoint sketch.

先在左边算第一问；不要把 $\mathbf{x}$ 的计算混进右侧中点图。

2. For the second equation, plot $U = (4, 0)$ and $W = (1, -2)$. Ask where Y must lie so that $U \rightarrow Y$ and $Y \rightarrow W$ are equal.

第二问先画 U 和 W ，问哪一点 Y 会使 $U \rightarrow Y$ 与 $Y \rightarrow W$ 相等。

3. Add Y at the midpoint. Label $U \rightarrow Y$ as $\mathbf{y} - \mathbf{u}$ and $Y \rightarrow W$ as $\mathbf{w} - \mathbf{y}$, then write $2\mathbf{y} = \mathbf{u} + \mathbf{w}$.

再标中点 Y ；两段箭头方向一致，分别标 $\mathbf{y} - \mathbf{u}$ 与 $\mathbf{w} - \mathbf{y}$ ，最后写 $2\mathbf{y} = \mathbf{u} + \mathbf{w}$ 。

Ask before explaining: What are $\mathbf{w}$ minus $\mathbf{y}$ and $\mathbf{y}$ minus $\mathbf{u}$ when $\mathbf{y}$ is five halves, minus one?

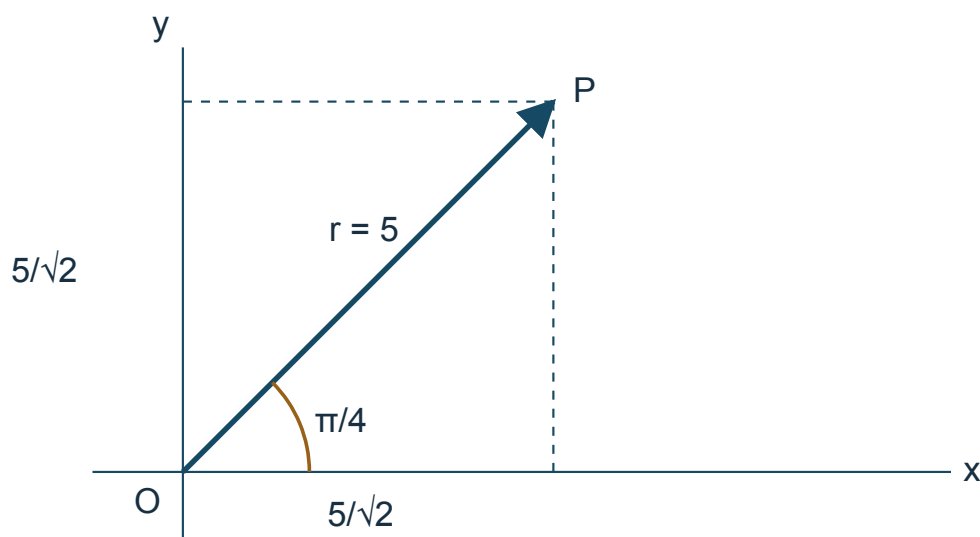
两边均为 $(-3/2, -1)$ ，验证成立；若不相等，先查负号。

A3 · Polar to Cartesian · 极坐标转直角坐标

Linear Algebra · Week 1 · 25 September 2026

TEACHER ONLY — contains solutions | 先让学生尝试，再按顺序画图与补答案

Find $\mathbf{x} \in \mathbb{R}^2$ such that $\mathbf{x} = \lambda \mathbf{u}(\theta)$ for $\lambda = 5$ and $\theta = \pi/4$. Note that $\mathbf{u}(\theta) = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}$.



First draw direction; then length; then project onto axes.

$$\mathbf{x} = 5(\cos(\pi/4), \sin(\pi/4)) = (5/\sqrt{2}, 5/\sqrt{2})$$

Board sequence · 板书顺序

1. Draw axes and a ray at $\pi/4$ in the first quadrant. Mark a length of 5 along the ray.
先画第一象限 $\pi/4$ 的射线，再标长度 5。
2. Drop perpendicular dashed lines to the axes; ask which component uses cosine and which uses sine.
向两轴画垂直虚线，问横、纵分量分别用什么。
3. Write $x=5 \cos(\pi/4)$, $y=5 \sin(\pi/4)$, then label both $5/\sqrt{2}$. Check the length is 5.
先写投影公式，学生算后再标两分量 $5/\sqrt{2}$ ；核对长度 5。

Ask before explaining: How do the components show both a length of five and an angle of pi over four?
平方和 25，所以长度 5；两分量正且相等，对应第一象限 45° 。

A4 · Cartesian to polar on the axes · 坐标轴上的极坐标

Linear Algebra · Week 1 · 25 September 2026

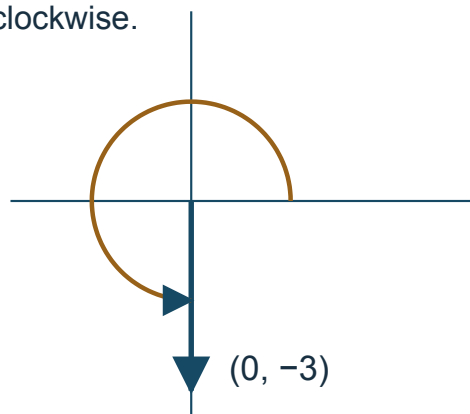
TEACHER ONLY — contains solutions | 先让学生尝试，再按顺序画图与补答案

Write $\begin{pmatrix} 3 \\ 0 \end{pmatrix}$ in the form $\mathbf{v} = \lambda \mathbf{u}(\theta)$ where $\lambda > 0$, $\theta \in [0, 2\pi)$, and $\mathbf{u}(\theta) = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}$. What about $\begin{pmatrix} 0 \\ -3 \end{pmatrix}$?

A positive radius; angles measured anticlockwise.



$$r = 3, \theta = 0$$



$$r = 3, \theta = 3\pi/2$$

$-\pi/2$ points down too, but is outside $[0, 2\pi)$.

$$(3, 0) = 3\mathbf{u}(0), \quad (0, -3) = 3\mathbf{u}(3\pi/2)$$

Board sequence · 板书顺序

1. Use two sketches. Draw (3,0) to the right and (0,-3) downwards, both from the origin.
左右两幅图，均从原点画箭头：向右与向下。
2. Keep the radius positive. Start angles at the positive horizontal axis.
半径保持正值；角度从正横轴开始量。
3. For the downward vector, trace an anticlockwise three-quarter turn and mark $3\pi/2$. Explain why $-\pi/2$ is excluded by this question.
向下的图画逆时针 3/4 圈，标 $3\pi/2$ ；解释 $-\pi/2$ 虽同方向却不在此题区间内。

Ask before explaining: Why is lambda equal to minus three not an allowed answer here?

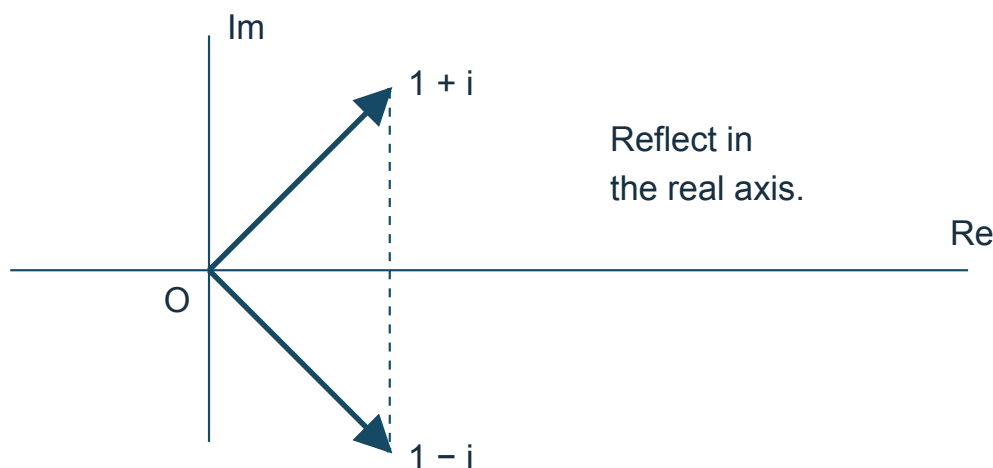
虽可配合另一角得到同一向量，但违反题目 $\lambda > 0$ ；长度必须为正。

A5 · Complex fraction · 复数分式

Linear Algebra · Week 1 · 25 September 2026

TEACHER ONLY — contains solutions | 先让学生尝试，再按顺序画图与补答案

Write $\frac{(3+2i)(3-2i)}{i+1}$ in the form $a+bi$ for $a, b \in \mathbb{R}$.



$$(1+i)(1-i) = 1^2 + 1^2 = 2$$

$$\frac{13}{1+i} = \frac{13(1-i)}{(1+i)(1-i)} = \frac{13}{2} - \frac{13}{2}i$$

Board sequence · 板书顺序

1. Draw $1+i$ and $1-i$ reflected in the real axis. This picture explains conjugation, not the full division algorithm.

画 $1+i$ 与 $1-i$ 关于实轴对称；这幅图解释共轭，不把图当完整除法证明。

2. On a separate line simplify the numerator: $(3+2i)(3-2i)=13$.

另起一行先把分子化为 13。

3. Write a bracket around BOTH numerator and denominator being multiplied by $1-i$. Then show the denominator is 2.

分子、分母旁同时标乘 $(1-i)$ ，强调两者都乘；再写分母为 2。

4. Finish with $13/2-(13/2)i$ and check by multiplying by $1+i$.

最后整理实部与虚部，再乘回 $1+i$ 检验。

Ask before explaining: Why is thirteen times one minus i not the final answer?

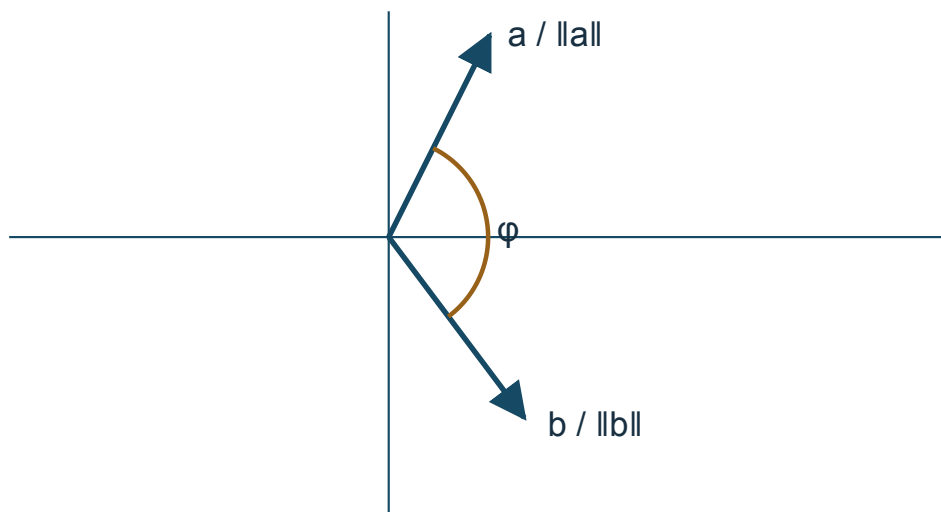
因为 $(1+i)(1-i)=2$ ，还必须除以 2；正确结果 $13/2-(13/2)i$ 。

A6 · Dot product and cosine · 点积与夹角余弦

Linear Algebra · Week 1 · 25 September 2026

TEACHER ONLY — contains solutions | 先让学生尝试，再按顺序画图与补答案

Compute the dot product $\begin{pmatrix} 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 6 \\ -8 \end{pmatrix}$ and hence determine the cosine of the angle between the vectors.



Unit directions shown: lengths rescaled only for this picture.

$a \cdot b = -10$, so $\cos \phi = -1/\sqrt{5} < 0$: the angle is obtuse.

$$a \cdot b = -10, \quad \|a\| = \sqrt{5}, \quad \|b\| = 10, \quad \cos \phi = -1/\sqrt{5}$$

Board sequence · 板书顺序

1. Only use this if students have met the dot product. Draw both directions from the same origin.

仅在已经学过点积时使用；两方向从同一原点画。

2. These arrows are unit directions, not the original vector lengths. Mark the smaller angle between them.

图中是单位方向，不代表原向量长度。标两方向之间的较小夹角。

3. Calculate with the original vectors: $a \cdot b = -10$, norms $\sqrt{5}$ and 10. Hence $\cos \phi = -1/\sqrt{5}$.

计算时仍用原向量：点积 -10 ，长度 $\sqrt{5}$ 与 10；得 $\cos \phi = -1/\sqrt{5}$ 。

Ask before explaining: Could minus ten be a cosine? What factor are we missing?

不可以，余弦必须在 $[-1, 1]$ ；缺少除以两模乘积 $10\sqrt{5}$ 。